

Measuring the gap between declared and real urgency

In a crisis, everyone shouts. SupplyScore is a digital twin that tells you **who is shouting, who has gone quiet, and which one should worry you.**

John HOARAU 484338 | R&D Intern, Direction Innovation Numérique, ALTEN | Individual Research Project 2025/2026

PANIC SPREADS, AND WE PUT A NUMBER ON IT

16×

When the shortage reached the headlines, managers whose own plants were untouched still raised their alarm level, about **sixteen times harder** than in a quiet week. That reflex is what floods premium freight and buries the alerts that matter.

Contagion effect, measured across 8 companies and 19 rounds.

IT FOUND THE BOTTLENECK, UNPROMPTED

7 of 10

Nobody told the model which of the eight companies mattered. Its shock simulation named the **semiconductor foundry**, the real chokepoint of the 2021 crisis, first, in seven of the ten rounds where the network was readable.

Systematic knock-out ranking, deep suppliers only.

IT PUTS THE RIGHT SUPPLIER ON TOP

3 in 4

Asked one round ahead which supplier is heading for trouble, it ranks the one that actually fails above one that does not, roughly **three times out of four**.

120 forward-looking forecasts scored against a frozen outcome.

1 THE BLIND SPOT Everything is green. The line stopped anyway.

A line stops. In the room where somebody should have escalated an hour earlier, every indicator on the wall is green.

Ask the team afterwards and you get the same two answers, in every industry.

Everything has been urgent for months, so nothing is. And **the thing that actually broke was the thing nobody flagged**, because the person closest to it believed they had it in hand, and had already escalated once too often to spend the credibility again.

Those are two different failures with opposite signs, and they do not cost the same. Panic wastes money and is recoverable. Silence is free, and it is not.

So the score weighs them differently: under-declaring a risk is penalised **2.25× harder** than over-declaring it, using the loss-aversion constant Kahneman and Tversky measured rather than a number chosen in-house.

The dashboard was not wrong. It was measuring the wrong thing.

2 THE OBJECTIVE Everyone scores the world. Nobody scores what we say about it.

Before building anything: surely someone measures this already. They do not.

The research literature models how disruption ripples across supplier tiers. The commercial platforms map multi-tier networks at a scale no prototype reaches. All of them score **physical reality**. None of them records what the human in charge actually declared, so none of them can compare the two.

That is the empty column, and filling it is the objective: elicit the declaration weekly without exhausting the operator, propagate it across the network, and measure the distance between what is said and what is true.

Everyone scores the world. Nobody scores what we say about it.

3 THE INSTRUMENT Two flows, one score

Real urgency comes from six families of operational indicators, treated as six smoke detectors. Any one of them saturating is enough to sound the alarm, because on a factory floor one bottleneck stops the whole line.

Averaging the six would hide the one case that matters. When a single family is at maximum and the other five are quiet, their average lands near **0.17**: on a ranked dashboard that supplier looks calm, while its line has in fact already stopped. Combining the six as independent alarms instead returns 1.00, and the supplier surfaces where it belongs.

Declared urgency comes from the manager, through a four-criterion weekly questionnaire that refuses contradictory answers. The two signals enter the chain at opposite ends and travel towards each other. **The distance between them is the score.**

One saturated detector has to trip the alarm. A mean would have buried it.

4 WHAT WAS BUILT A score you can argue with

Put a number on somebody's judgement and they will contest it, usually within ten minutes. A score that cannot answer back does not survive its first meeting.

So every score **decomposes exactly**, criterion by criterion and block by block, with no residual left over, and every change carries its own record: who entered it, when, and under which rule. Dispute a 46.5 and the tool replies term by term, on screen, with the trail underneath.

SupplyScore runs today and is still under active development: a live dashboard, a weekly review cycle, network simulation, and an explanation page behind every figure.

A score nobody can interrogate is a score nobody uses.

5 THE TEST HÉLIOS: the chip crisis, replayed blind

To test it, the 2020–2022 semiconductor shortage was rebuilt as an eight-company, eighteen-round serious game, and the players were never told which crisis they were living through.

Authentic public series feed the physical side, alongside twelve calibrated events pinned to dated real ones: the Texas freeze, the Renesas fire, the Suez blockage, the Taiwan drought, the Malaysian back-end shutdowns, the autumn bullwhip.

Eight managers, one programme, and a chain that tightens round by round. Each of them sees only their own corner of it, and each has to say every week how worried they are.

The crisis was real. Only the names were changed.

6 WHAT IT FOUND, AND WHERE IT GOES Perception became a measured quantity

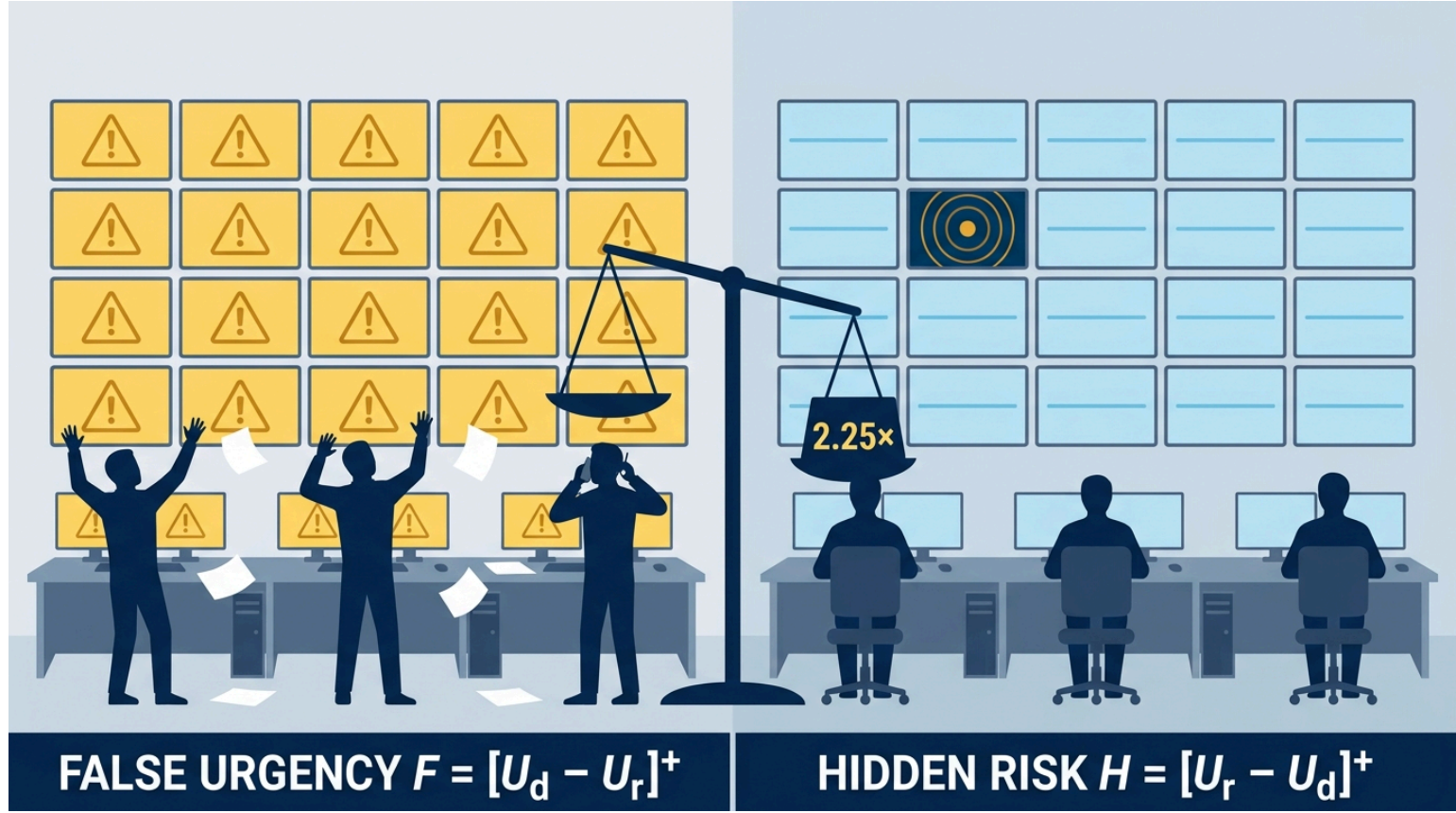
The campaign produced the first evidence that the gap is measurable, and that it carries information nobody else is collecting.

Contagion is the headline. Managers with nothing physically wrong at their own site still escalated on the rounds when the crisis made the news. Alert inflation, caught in the act and given a number for the first time.

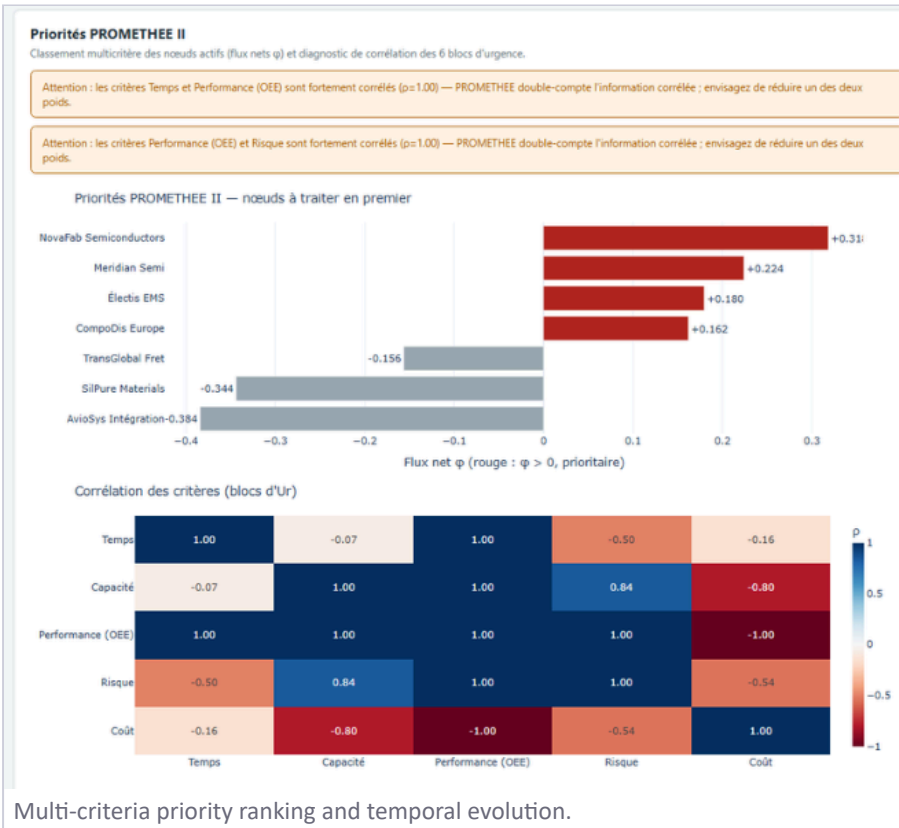
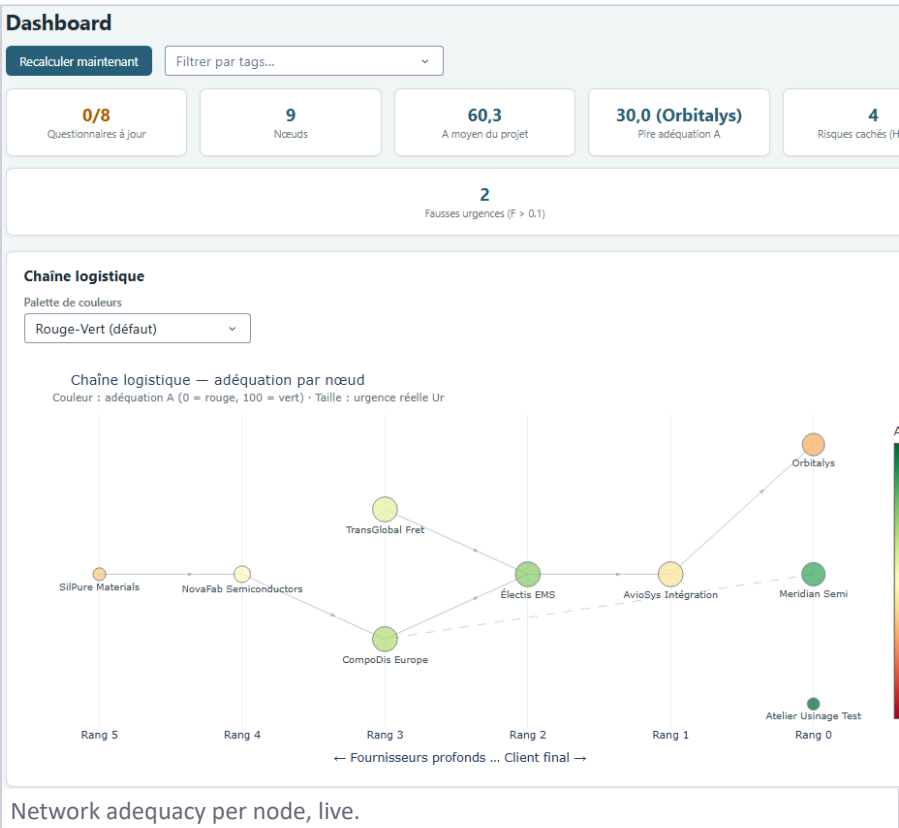
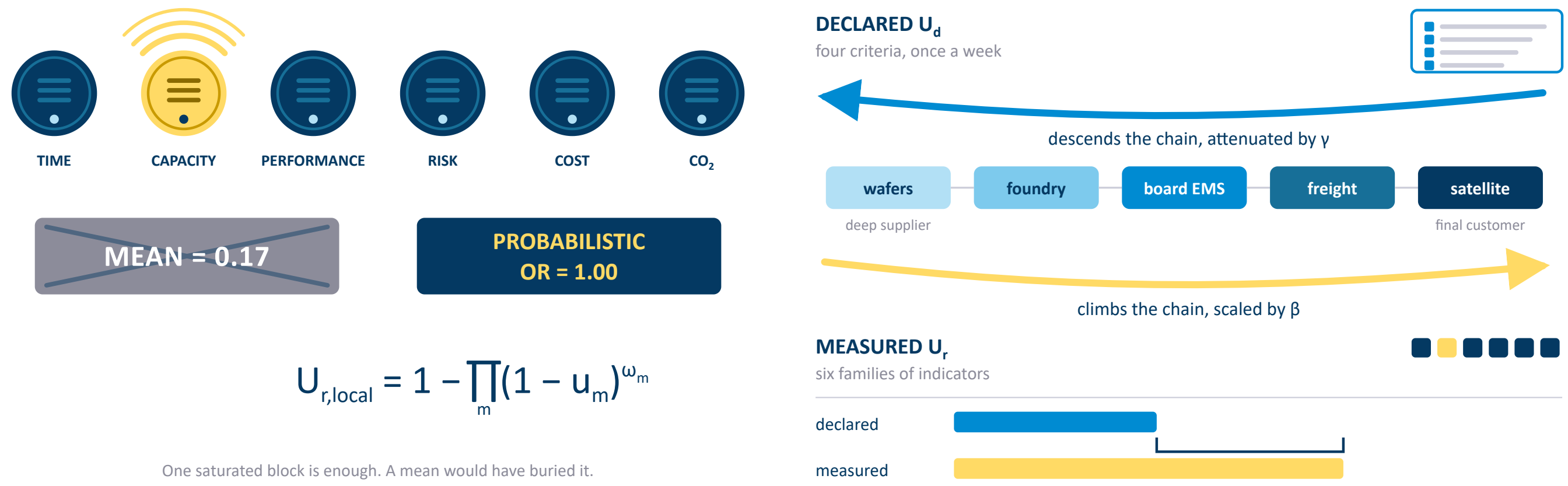
The network layer found the chokepoint on its own, and the forecast layer ranks the right supplier first three times in four, one round ahead.

Eighteen rounds were enough to show the signal exists. What it needs now is volume: more campaigns, on more networks and more crises, each one sharpening the next.

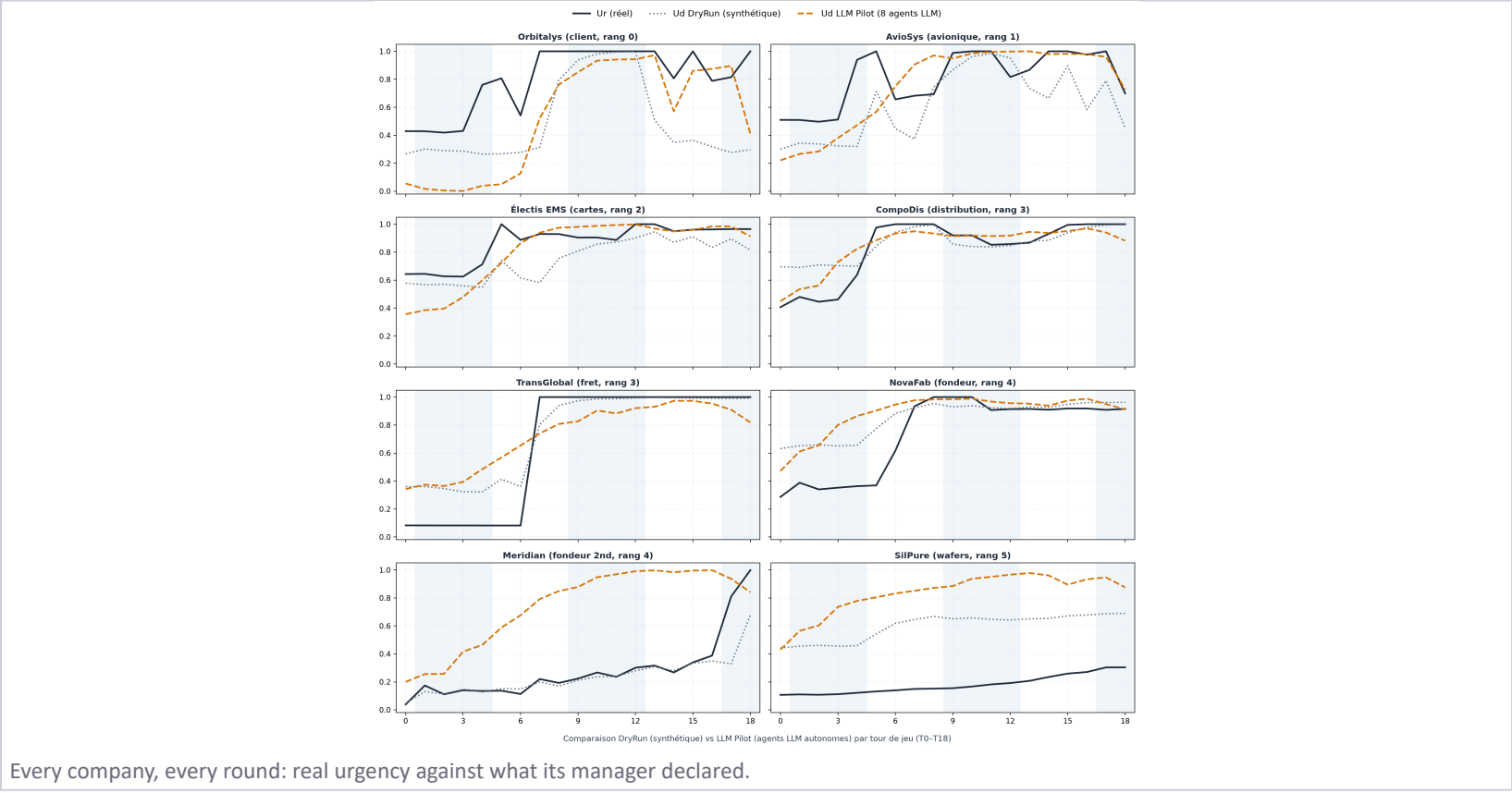
Perception is now a measured quantity, not a mood.



	Ripple-effect models	Markov and dyn. Bayesian	Bayesian SC risk networks	TTR / TTS stress tests	Everstream, Resilinc, Interos	SupplyScore
Scores physical reality	✓	✓	✓	✓	✓	✓
Scores human declaration	✗	✗	✗	✗	✗	✓
Scores the gap	✗	✗	✗	✗	✗	✓



Détail des nœuds						
Fond rouge clair : A < 40 · H en rouge : H > 0.3 · Hébo coloré selon le statut (A/F/H) = — = si questionnaire manquant.						
iNom	iRang	iStatut	iHebo	iUd_loc	iUr_loc	iUd
filter data...						
Atelier Usage Test	0	Active	Manquant	0	0	0
Meridian Semi	0	Active	En retard (1 sem.)	0.841	0.879	0.841
Oribitals	0	Terminée	En retard (1 sem.)	0.409	0	0.409
AvidSys Integration	1	Active	En retard (1 sem.)	0.568	1	0.727
Electis EMS	2	Active	En retard (1 sem.)	0.793	1	0.914
CompoDis Europe	3	Active	En retard (1 sem.)	0.561	1	0.882
TransGlobal Fret	3	Active	En retard (1 sem.)	0.666	1	0.819
Novafab Semiconductors	4	Active	En retard (1 sem.)	0.575	0.325	0.912
SiPure Materials	5	Active	En retard (1 sem.)	0.656	0.048	0.876



Contagion, quantified

Untouched companies escalated **16×** harder on news rounds than on quiet ones. The pre-registered prediction held.

Bottleneck, unprompted

The foundry ranked first among deep suppliers in **7 of 10** readable rounds, with no hint given.

Ranking that works

3 in 4 forecasts put the supplier that failed above one that did not, a round ahead.

NEXT

1. **Run more campaigns**, on other networks and other crises, and recalibrate the model on the data each one produces.
2. **Fuse the second brick**: warehouse volumetry read from satellite imagery, so capacity is observed rather than declared.
3. **The end goal: a language model fed by these forecasts**, answering in the moment rather than once a week, and stating what is about to happen with the probability attached.